# Predicting Rain Tomorrow in Australia

An explainable ML model for travel planning decisions

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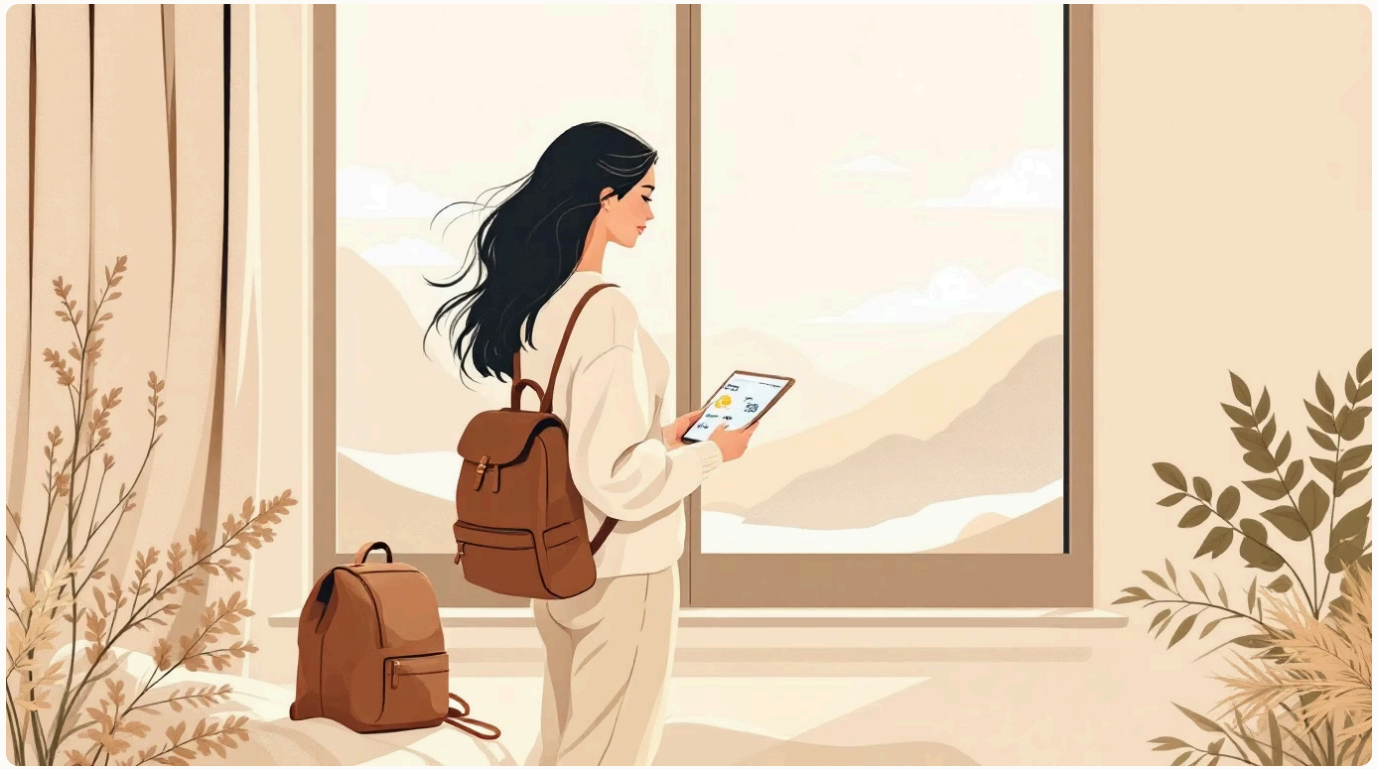
# The Traveler's Question

## The Core Question

"Will it rain tomorrow?" (Yes/No) — a binary prediction that shapes travel plans

## Why Travelers Care

- Planning outdoor activities and routes
- Packing appropriate gear and clothing
- Avoiding costly cancellations
- Optimizing time and budget



## Model Output

Probability of rain + alert threshold triggers

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# Dataset Snapshot

**126k**

Daily Observations

Rows of weather data

**11**

Features

Variables including target

**78%**

No Rain

Majority class

**22%**

Rain Tomorrow

Minority class

Source: "Rain in Australia" Kaggle dataset — daily weather observations across Australian weather stations

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# Feature Engineering Choices

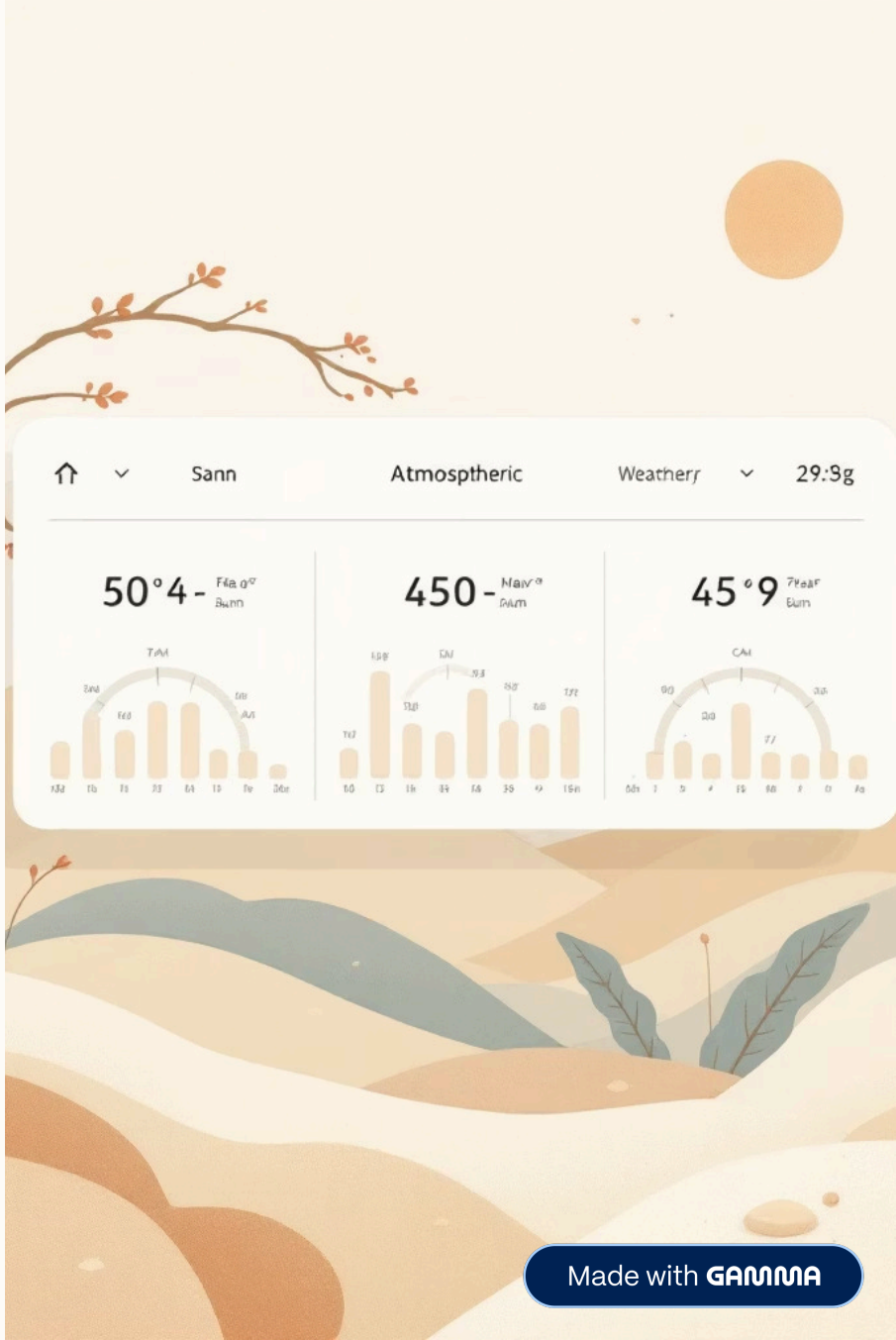
## Features We Used

- Atmospheric signals: humidity, pressure, cloud cover
- Wind speed and direction
- Temperature readings (average)
- Daylight: sunshine and solar exposure
- Seasonality: month indicator

## Features We Excluded

- **RainToday/Rainfall:** Persistence shortcuts excluded on purpose
- **Location:** Removed to train a general model

*Trade-off: May reduce accuracy but improves generality across locations*

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# Data Preparation Process

01

## Missing Values

Pragmatic handling to keep dataset usable

02

## Feature Engineering

Combine 9am/3pm readings into daily averages + month seasonality

03

## Train/Test Split

Random stratified (75/25) to preserve class balance

04

## SMOTE Application

Synthetic oversampling applied on training set only (not test)

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# How We Measure Success

## Why Accuracy Misleads

With 78% No Rain class, a model that always predicts "No Rain" achieves 78% accuracy — useless for travel planning

## Primary Metrics

- **Balanced Accuracy:** Treats both classes fairly
- **Macro F1:** Balances precision and recall

## Goal

Improve recall for "Rain=Yes" while controlling false alarms

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# Three Models Compared

### Logistic Regression

Interpretable baseline

### Random Forest

Non-linear baseline

### Gradient Boosting

Strong tabular model

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# Best Model Performance

## Model Summary

**Winner model:** Random Forest

78%

Balanced Accuracy

72%

Macro F1

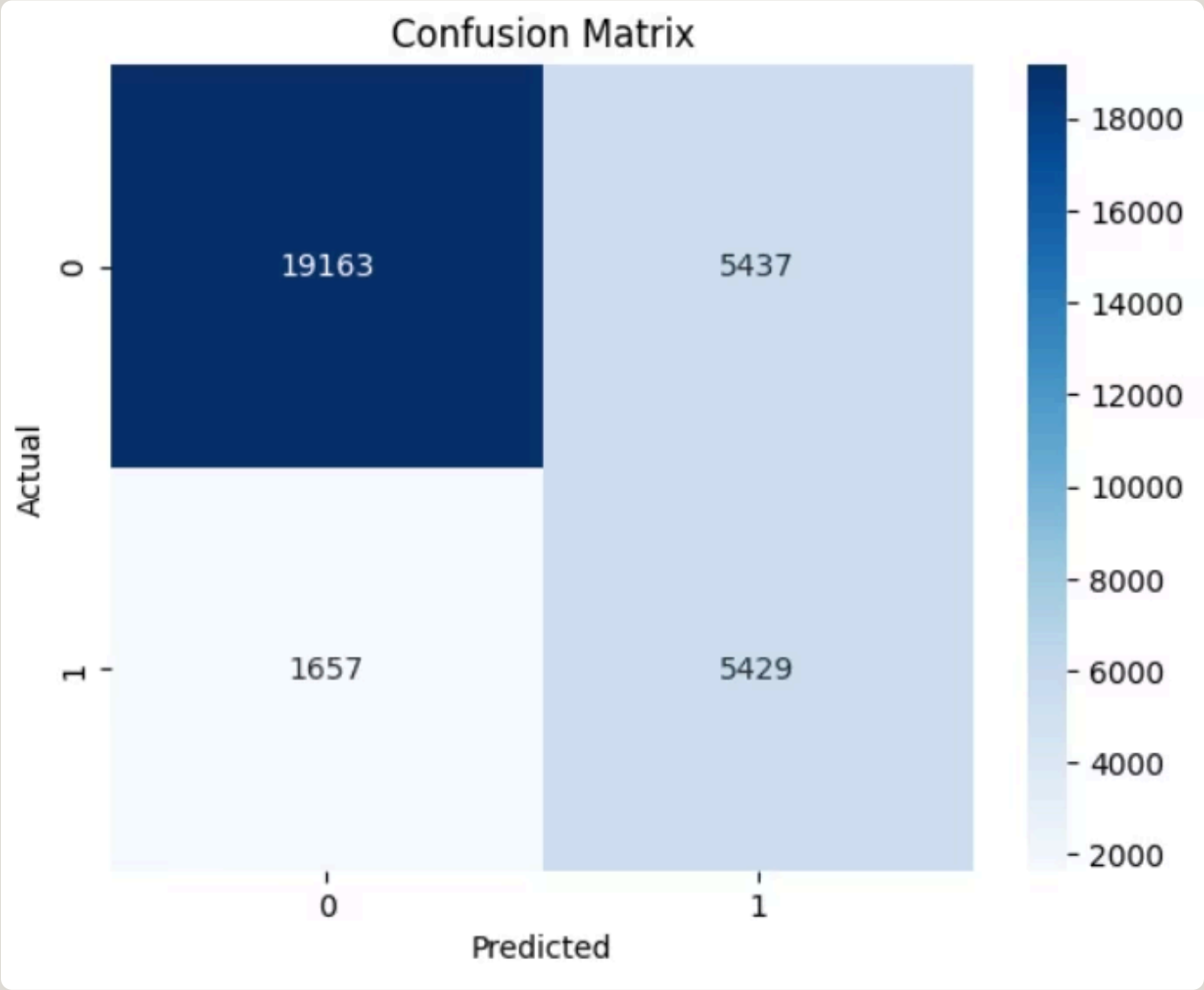
77%

Macro Recall

## Key Metric

**Minority-class recall (Rain=Yes): 77%**

Critical for travel alerts — catching rain days matters most

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# What Drives Rain Risk?

## SHAP Explainability Analysis

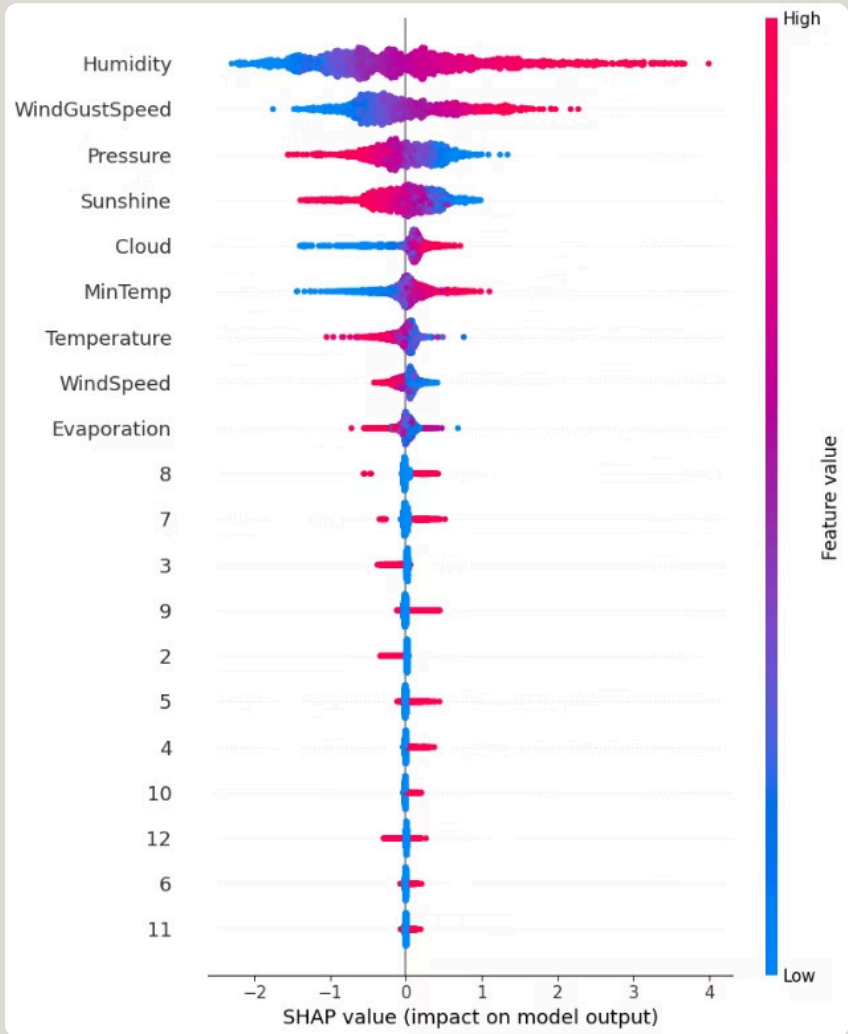
Feature importance from SHAP summary plot:

### Higher Rain Probability

- Higher humidity
- More cloud cover
- Stronger wind gusts

### Lower Rain Probability

- Higher pressure
- More sunshine
- Clearer skies

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# Limitations & Next Steps

## Current Limitations

- Random split (not time-based)
- Location removed
- Pragmatic missing-value handling

## Future Improvements

- Time-based validation
- Compare "with vs without persistence features"
- Calibrate probabilities

> An explainable rain-risk predictor that supports better travel planning decisions

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